Essential Terminal, Git & Cloud Computing Digital Skills Handbo

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1. Essential Terminal Commands (Bash / Zsh)

- Navigation: cd, ls -la, pwd, mkdir -p, and rm -rf (use with extreme caution).
- File inspection & editing: cat, less, head, tail -f, grep -r "pattern", and nano/vim.
- Permissions & Environment: chmod +x script.sh, export PORT=3000, and env inspection.

2. Git Version Control Workflows

- Standard Workflow: git status -> git add . -> git commit -m "feat: descriptive message" -> git push.
- Branching Strategy: git checkout -b feature/branch-name; never develop directly on production main.
- Resolving Merge Conflicts: Inspect conflict markers, verify code logic, stage resolved files, and commit.

3. Cloud Hosting & Serverless Deployment

- Static Hosting vs VPS: When to use Netlify/Vercel (Jamstack) vs Docker containers on Linux VPS.
- Environment Variables: Never commit .env files containing API keys or private credentials to public git repos.
- Automated CI/CD Pipelines: Continuous integration triggers automated builds and tests upon each git push.